

Probing the transitions between 2D and 3D states in InGaN/GaN quantum wells via admittance spectroscopy

Cite as: Appl. Phys. Lett. **128**, 173301 (2026); doi: 10.1063/5.0324742

Submitted: 26 January 2026 · Accepted: 31 March 2026 ·

Published Online: 27 April 2026



View Online



Export Citation



CrossMark

Zilan Wang,^{1,2,a)} Xue-Qi Wang,² Yuning Yi,¹ Haoyang Li,¹ Bo Liu,³ Francis Chi-Chung Ling,² and Lai Wang^{3,a)}

AFFILIATIONS

¹School of Optoelectronic Engineering and Instrumentation Science, Dalian University of Technology, Dalian 116024, China

²Department of Physics, The University of Hong Kong, Hong Kong, China

³State Key Laboratory of Widegap Semiconductor Optoelectronic Materials and Technologies, Department of Electronic Engineering, Tsinghua University, Beijing 100084, China

^{a)}Authors to whom correspondence should be addressed: wangzilan@dlut.edu.cn and wanglai@tsinghua.edu.cn

ABSTRACT

Carrier transport and recombination mechanisms in InGaN/GaN multiple quantum well (MQW) structures are fundamental to the performance of advanced optoelectronic devices. Thermally activated signals with activation energies of 60–100 meV are frequently observed, yet their physical origin remains a subject of debate between point defects and interface states. In this work, we provide a new perspective on these signals through systematic admittance spectroscopy analysis of MQW p-n junction diodes. Experimental results demonstrate that the observed electrical response originates from the intrinsic kinetic process of carrier emission from 2-dimensional (2D) bound states in the quantum wells to 3-dimensional (3D) free states. The signal intensity exhibits a pronounced bias dependence, which correlates directly with the number of quantum wells intersecting the Fermi level, a behavior fundamentally distinct from that of bulk defects. Furthermore, the activation energy remains nearly constant under varying bias, ruling out the interface-state hypothesis where energy levels typically shift with surface potential. By employing a 2D-to-3D emission model, we quantitatively extract the carrier capture velocity and associated kinetic parameters. This study not only helps clarify the long-standing controversy surrounding low-energy signals in InGaN/GaN systems but also provides essential insights into carrier accumulation and transport dynamics for the optimization of MQW-based optoelectronic devices.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0324742>

InGaN/GaN multiple quantum wells (MQWs) constitute the core active region of a wide range of GaN-based optoelectronic devices, including photodetectors, high-electron-mobility transistors, light-emitting diodes, laser diodes, and advanced display technologies.^{1–5} The performance of these devices is critically governed not only by band structure and radiative recombination efficiency but also by carrier transport and capture-emission dynamics occurring within and around the QWs. In particular, low-energy carrier processes on the sub-0.1 eV scale can significantly influence response speed, internal quantum efficiency, and noise characteristics, yet remain challenging to unambiguously interpret.

Owing to quantum confinement, carriers in MQW structures may occupy either 3-dimensional (3D) extended states in the barrier layers or 2-dimensional (2D) bound states within the QWs. Carrier

transitions between these states introduce distinct thermal activation behaviors that differ fundamentally from those associated with conventional bulk point defects. While optical spectroscopies such as photoluminescence and time-resolved measurements have provided valuable insight into recombination and relaxation processes in MQWs,⁶ electrical characterization techniques offer complementary advantages for probing low-energy transport phenomena and kinetic parameters that directly govern device operation.

Deep-level transient spectroscopy (DLTS) has frequently revealed thermally activated signals with activation energies in the range of 60–100 meV in GaN-based devices.^{7,8} The physical origin of this response, however, has remained controversial. Early interpretations attributed these signals to nitrogen vacancies or related bulk defects,⁹ despite first-principles calculations predicting transition

levels far from this energy range and experimental evidence indicating that such vacancies exhibit high formation energies and are unlikely to form spontaneously in n-type GaN.^{10,11} Alternative explanations have proposed interface-related states, including donor-like traps located tens of meV above the valence band at negatively charged InGaN/GaN interfaces.¹² Nevertheless, DLTS intrinsically involves transient depletion-width modulation and can probe both majority and minority carriers,¹³ making it difficult to precisely identify the microscopic origin of low-energy signals—especially in structures containing MQWs.

The carrier escape and capture kinetics in quantum wells (QWs) are often more complex than the classical thermionic emission models suggest. As demonstrated by Zubkov *et al.*,¹⁴ the transition from 2D bound states to 3D continuous states can be mediated by quasi-bound or quasi-resonant states located above the QW. These intermediate states provide a critical pathway for resonant tunneling and dimensionality-dependent transport, distinguishing QW carrier dynamics from the behavior of isolated point defects in bulk semiconductors.

Admittance spectroscopy (AS) provides a powerful steady-state alternative for investigating such carrier dynamics.^{15–18} By measuring frequency- and temperature-dependent conductance and capacitance under a fixed bias, AS predominantly probes majority-carrier processes. The response originates from the carrier emission and capture dynamics that occur when the Fermi level crosses the energy levels of the active states. Compared with DLTS, this approach avoids complications associated with transient bias switching and depletion-width modulation, enabling more reliable identification of low-energy states and extraction of kinetic parameters such as capture velocity.

In this work, we employ AS to investigate the electrical response of InGaN/GaN MQW p-n junction devices. We demonstrate that a commonly observed low-energy admittance signal originates from carrier emission between 2D bound states in the QWs and 3D free states, rather than from conventional point defects or interface traps. Furthermore, the capture velocity associated with this process and its bias dependence are quantitatively extracted, providing direct insight into the kinetic role of QWs in GaN-based optoelectronic devices.

The sample parameters of the InGaN/GaN MQWs can be referred to our previous work.¹⁹ AS was carried out using an impedance analyzer (MFIA, Zurich Instruments) with an oscillation voltage of 30 mV across a frequency range from 1 kHz to 1 MHz. The sample temperature was controlled by a Lakeshore Model 336 temperature controller, spanning from 10 to 300 K at a constant ramp rate of 1 K/min. The bias ranges from -1 to $+0.5$ V.

Figure 1 presents the simulated energy band diagram of the MQW device at 0 V bias, calculated by the commercial software SiLENSe. The band bending of the p-n junction and the position of the Fermi level can be clearly observed. The MQW active region is unintentionally doped and exhibits weak n-type conductivity. Given the high hole concentration in the p+-GaN layer (10^{19} cm^{-3}), the junction forms an asymmetric p+-n structure, where the depletion region extends primarily into the n-type MQW. In admittance spectroscopy, the measured signal originates from the majority carrier response within the depletion region. Due to the high doping contrast, the junction capacitance modulation is dominated by the depletion region extension into the MQW, effectively filtering out the hole contribution from the p-side.

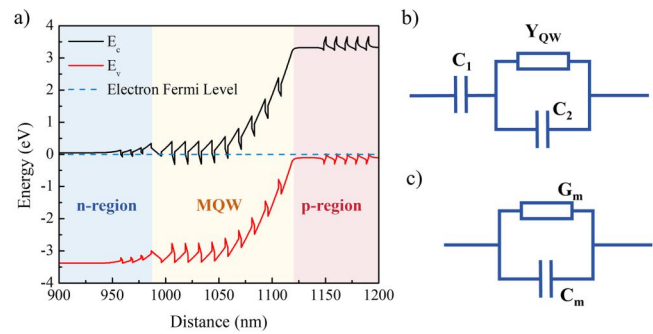


FIG. 1. (a) Energy band diagram of a 9-period InGaN/GaN MQWs device under 0 V bias. The horizontal axis represents the relative position, and the vertical axis corresponds to the energy scale. The Fermi level is indicated by the blue line in the diagram. (b) Equivalent electrical circuit of the InGaN/GaN LED structure. (c) Real admittance component measurements in the parallel circuit mode.

According to the principle of AS, an electrical response occurs only at intersections between the Fermi level and trap energy levels. Based on the band diagram, it is most likely that the intersections between the Fermi level and the 3–4 MQWs on the left side are responsible for the observed response.

To clarify the contribution of MQWs to the total impedance, we employ an equivalent electrical circuit model [Fig. 1(b)] for the p-n junction. By incorporating the QW admittance as a parallel G_{QW} - C_{QW} network within the space-charge region, the total series impedance Z_s is expressed as²⁰

$$Z_s = \frac{1}{j\omega C_1} + \frac{1}{G_{QW} + j\omega(C_2 + C_{QW})}. \quad (1)$$

Here, G_{QW} and C_{QW} represent the conductance and capacitance contributions arising from the QWs, respectively. C_1 denotes the geometric capacitance, while C_2 corresponds to the capacitance of the space-charge region. We measure impedance (Z_m) circuit

$$Z_m = \frac{1}{G_m + j\omega C_m}, \quad (2)$$

here, G_m and C_m represent the measured conductance and capacitance, respectively, as shown in Fig. 1(c).

Let $Z_s = Z_m$ and equate their real and imaginary parts, respectively. Then, the value of G_{QW} is deduced

$$\frac{G_{QW}(\omega)}{\omega} = \frac{(G_m C_1^2 \omega)}{G_m^2 + (C_m - C_1)^2 \omega^2}, \quad (3)$$

where C_1 can be determined from the capacitance measured under high-frequency conditions.

To describe the carrier exchange between the 2D quantum well (QW) and the 3D barrier states, we employ a phenomenological rate equation approach. Although a complete kinetic treatment would require the Boltzmann transport equation (BTE), Eq. (4) effectively captures the dominant transition dynamics under small-signal ac modulation,

$$\frac{G_{QW}(\omega)}{\omega} = C_T \frac{\omega \tau}{1 + \omega^2 \tau^2}, \quad (4)$$

where C_T is the capacitance attributed to carrier trapping and detrapping processes within the QWs.

In this framework, the thermal emission time constant (τ) is treated as an integrated response of both occupation statistics and transition probabilities,

$$\tau = \frac{AN_{eq}}{c_n n_{eq}}. \quad (5)$$

Here, A is the electrode area and c_n denotes the electron capture coefficient. N_{eq} is the 2D carrier density in the well in steady state,

$$N_{eq} = \left(\frac{m^*}{\pi \hbar^2} \right) kT \exp \left(\frac{E_F + qV_{QW} - E_1}{kT} \right). \quad (6)$$

The temperature-dependent exponential term accounts for the Boltzmann-tail distribution of carriers. Under the bias conditions, where the Fermi level (E_F) is near the subbands, this term represents the probability of carriers occupying high-energy states capable of escaping the well.

E_1 denotes the bound energy level in QW. n_{eq} is 3D density above the well. By defining

$$v_{QW} = \frac{c_n}{A}, \quad (7)$$

$$n_{eq} = 2 \left(\frac{m^*}{2\pi \hbar^2} \right)^{\frac{3}{2}} (kT)^{\frac{3}{2}} \exp \left(\frac{E_F + qV_{QW} - E_2}{kT} \right). \quad (8)$$

Then, substituting it into Eq. (5),

$$\tau = \frac{1}{v_{QW}} \left(\frac{2\pi \hbar^2}{m^* kT} \right)^{\frac{1}{2}} \exp \left(\frac{E_2 - E_1}{kT} \right). \quad (9)$$

The effective capture/emission velocity (v_{QW}) reflects the quantum mechanical transition probability,²¹ which is governed by the wavefunction overlap between the 2D bound states and the 3D (or quasi-bound) states.

m^* is the effective electron mass, $\hbar$ is the reduced Planck constant, and E_1 is the bound energy level. Owing to the polarization-induced band tilting, E_2 acts as a quasi-bound state that aligns with the barrier continuum, facilitating rapid carrier escape. The time constant τ , the energy level depth $E_2 - E_1$, and the QW capture velocity v_{QW} can then be calculated accordingly.

Figure 2 shows the normalized conductance (G_{QW}) as a function of frequency under 0 V bias, measured at different temperatures. The emergence of the conductance peak signifies a frequency-modulated response governed by carrier capture and emission kinetics between the 2D bound states in the QW and the 3D continuum. The dashed line represents the theoretical fit based on Eq. (4), from which the characteristic time constant τ is extracted to construct the Arrhenius plot across the investigated temperature range. While the model yields excellent agreement with experimental data at lower frequencies, a noticeable broadening is observed on the high-frequency side. This deviation suggests that the carrier emission from a 2D quantum-confined system possesses a more complex kinetic nature than that of isolated point defects in bulk semiconductors, likely due to the dimensionality-dependent density of states transition. A detailed line shape analysis of the conductance peaks shown in Fig. 2(a) is provided in the [supplementary material](#).

From the Arrhenius fitting in Fig. 2(b), the energy level difference $E_2 - E_1$ is determined to be approximately 75 meV, with a capture velocity of about 9.878×10^7 cm/s. This demonstrates excellent linear behavior, indicating that the experimental results are consistent with the description in Eq. (9). The results obtained at various voltages all exhibit excellent linear fitting, indicating that the physical process is dominated by thermal activation. If the point defect model from AS¹³ is used for explanation, the goodness of fit of the linear regression decreases significantly, and the estimated energy level is approximately 62 meV. However, more importantly, subsequent results further confirmed that the electrical response does not originate from point defects.

Figures 3(a)–3(d) show the AS results measured at four bias voltages: -1 , -0.5 , 0 , and $+0.5$ V. While the peak shapes and their corresponding characteristic frequencies remain almost unchanged, the peak intensities vary significantly. By applying the Arrhenius fitting method described in Fig. 2, the activation energy, capture velocity, and trap concentration ($N_T = 2N_D C_T/C_1$, where N_D is the carrier concentration) corresponding to each voltage can be extracted, as summarized in Table I. As the applied bias voltage increases, the signal intensity increases notably, manifesting a behavior distinct from typical point defect levels. This trend should be attributed to the number of QWs involved in the trapping process.

Figure 4(a) presents the simulated energy band diagrams of the MQWs structure under bias voltages ranging from -1 to $+0.5$ V. At

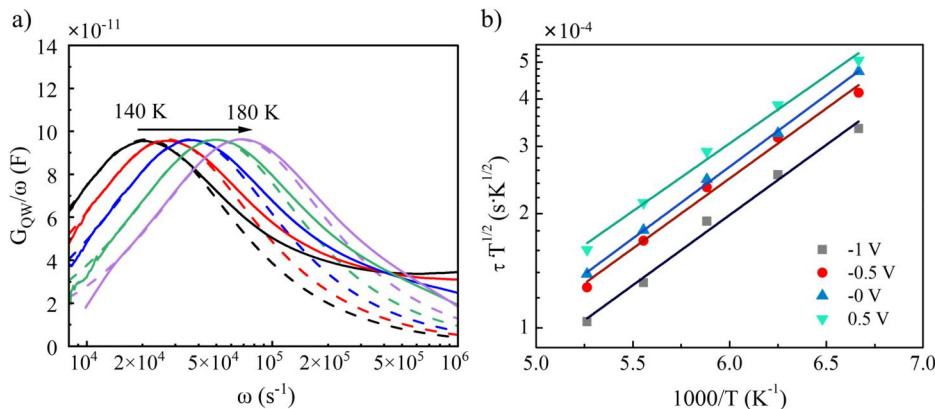


FIG. 2. (a) Experimental frequency-normalized G_{QW} as a function of angular frequency ω , with theoretical fitting results shown by the dashed line. (b) Corresponding Arrhenius plot.

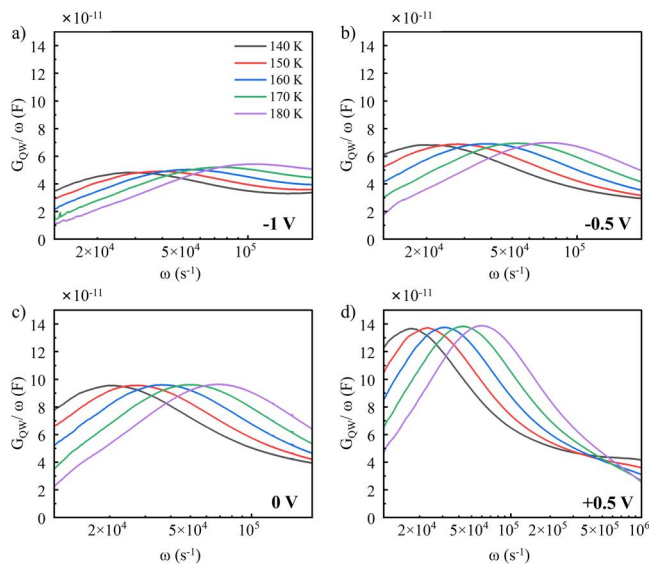


FIG. 3. AS results measured under bias voltages from -1 to $+0.5$ V: (a) -1 , (b) -0.5 , (c) 0 , and (d) $+0.5$ V.

TABLE I. Capture velocity and E_2-E_1 energy values extracted from AS measurements under bias voltages ranging from -1 to $+0.5$ V.

Bias (V)	E_2-E_1 (meV)	Capture velocity (10^7 cm/s)	Concentration N_T (10^{16} cm $^{-3}$)
-1 V	73 ± 4	11.8	2.2
-0.5 V	73 ± 3	9.3	3.1
0 V	75 ± 2	9.9	4.3
0.5 V	71 ± 3	6.4	6.23

-1 V, the depletion region is widest, resulting in only three QWs fully lying below the Fermi level. As the voltage increases to $+0.5$ V, the number of QWs below the Fermi level rises to approximately five, as indicated by the blue shaded regions in the diagram. The clear voltage-dependent variation in the AS signal intensity observed in

Fig. 3 strongly correlates with the number of QWs involved, as further illustrated in Fig. 4(d). This phenomenon further confirms that the observed response does not originate from conventional point defect levels, as the concentration of such defects would not vary with applied voltage.

We also observed this trapping level in DLTS measurements, located at approximately 83 meV, consistent with the results obtained from AS. Previous studies attributed this signal to interface states, based on evidence from current DLTS measurements, showing that the traps were located at the InGaN/GaN interface rather than in the bulk, along with noticeable shifts in the CV curves. However, these observations are also fully consistent with the conclusions of this paper: the trapping behavior of 2D traps resembles that of interface states and occurs only within a few-nanometer-wide QWs rather than bulk traps.

The discrepancy with the interface-state hypothesis lies in the fact that the energy level observed here remains almost unchanged under different voltages, whereas interface-state energies are typically strongly correlated with surface potential variations induced by voltage. More importantly, previously reported interface states were located about 60 meV above the valence band,¹² a region inaccessible to the AS measurements in this study, which are sensitive only to majority-carrier responses. Band diagram analysis further confirms that the Fermi level cannot reach this position. Therefore, the possibility of electrical response observed in this work originating from interface states could be ruled out.

It is important to note that in InGaN/GaN heterostructures, the presence of strong internal polarization fields leads to a significant tilting of the band diagram. This effect transforms the potential barriers into an asymmetric triangular shape, which contrasts with the conventional rectangular barriers found in many other III-V materials. Under such conditions, the carrier escape from 2D bound states to the 3D continuum follows a more complex, non-reciprocal path. The 75 meV activation energy extracted in this work reflects the effective barrier height in this distorted potential landscape, where the dimensionality transition is further modulated by the spatial separation of wavefunctions and the polarization-induced barrier lowering.

In summary, this work systematically investigates a low-energy electrical response commonly observed in InGaN/GaN MQWs structures using AS. Rather than attributing this response to conventional bulk defects or interface states, we demonstrate that it originates from carrier capture and emission between 3D transport states and 2D bound levels within the QWs. This process reflects an intrinsic kinetic

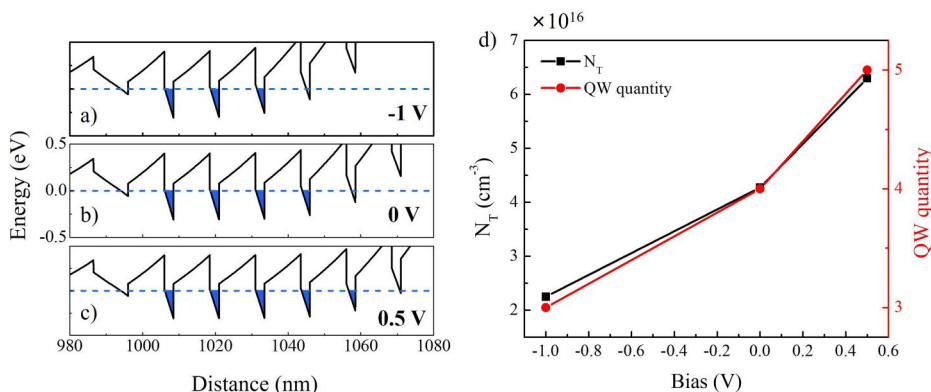


FIG. 4. Schematic energy band diagrams of the MQW structure showing the Fermi level positions under (a) -1 , (b) 0 , and (c) $+0.5$ V. (d) Illustration of the electrically responsive carrier concentration as a function of the applied bias voltage.

feature of MQW structures and can easily be misinterpreted if analyzed solely within the framework of point defect models.

By employing AS, we effectively eliminate ambiguities associated with depletion-width modulation and minority-carrier contributions, enabling reliable extraction of the activation energy and capture velocity associated with the QW states. The observed bias dependence of the signal intensity further confirms its direct correlation with the number of QWs participating in the capture process, a behavior incompatible with conventional defect-related responses.

The identification and quantification of the QW capture velocity provide important insight into carrier transport dynamics in GaN-based optoelectronic devices. This kinetic parameter directly governs the coupling strength between 2D bound states and 3D free carriers, thereby influencing response speed, carrier accumulation, and recombination behavior in photodetectors and light-emitting devices. Beyond clarifying the long-standing controversy surrounding low-energy admittance and DLTS signals in InGaN/GaN systems, the present results highlight the necessity of incorporating QW capture dynamics into the electrical analysis and performance optimization of GaN-based MQW devices.

See the [supplementary material](#) for detailed fitting parameters for the conductance peaks in [Fig. 2\(a\)](#) (Table S1) and a comprehensive line shape analysis ([Fig. S2](#)).

All the authors gratefully acknowledge the National Key Research and Development Program (No. 2021YFA0716400), the National Natural Science Foundation of China (Nos. 62374023, 62204035, 62225405, 62350002, 61991443, 62127814, 62235005, and 61927811), the Key R&D Project of Jiangsu Province, China (No. BE2020004), and the Collaborative Innovation Centre of Solid-State Lighting and Energy-Saving Electronics.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Zilan Wang and Xue-Qi Wang contributed equally to this paper.

Zilan Wang: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal). **Xue-Qi Wang:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Writing – review & editing (equal). **Yuning Yi:** Data curation (equal). **Haoyang Li:** Data curation (equal). **Bo Liu:** Data curation (equal); Writing – review & editing (equal). **Francis Chi-Chung Ling:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Supervision (equal); Writing – review & editing (equal). **Lai Wang:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

- ¹R. Zhang *et al.*, “Recent progress in GaN-based ultraviolet photodetectors,” *J. Mater. Chem. C* **13**(22), 10972–10996 (2025).
- ²U. K. Mishra, P. Parikh, and Y. F. Wu, “AlGaIn/GaN HEMTs—an overview of device operation and applications,” *Proc. IEEE* **90**, 1022 (2002).
- ³M. Buffolo *et al.*, “Defects and reliability of GaN-based LEDs: Review and perspectives,” *Phys. Status Solidi (a)* **219**(8), 2100727 (2022).
- ⁴Z. Liu, D. Wang, H. Gao, M. Li, H. Zhou, and C. Zhang, “Metasurface-enabled augmented reality display: A review,” *Adv. Photonics* **5**, 034001 (2023).
- ⁵Z. Li, Z. Xu, S. Li, X. Zhang, Y. Zhou, J. Wang, Z. Hao, Y. Luo, C. Sun, B. Xiong, Y. Han, and L. Wang, “21.79/17.49 Gbps full-duplex visible light communication enabled by short-cavity blue and green InGaIn/GaN laser diodes,” *J. Lightwave Technol.* **43**, 3348 (2025).
- ⁶B. Liu, C. Sun, L. Yu, Z. Hao, Y. Luo, C. Sun, B. Xiong, Y. Han, J. Wang, H. Li, L. Gan, and L. Wang, “Inverted V-shaped size-dependent emission wavelength shift in InGaIn micro-LEDs with size down to 1 μm ,” *Appl. Phys. Lett.* **125**(9), 091103 (2024).
- ⁷S. Mohanty *et al.*, “Design of ultra-scaled-channel N-polar GaN HEMTs with high charge density: A systematic study of hole traps and their impact on charge density in the channel,” *J. Appl. Phys.* **128**, 235701 (2020).
- ⁸R. Lingapathi *et al.*, “Source of 2D electron gas in unintentionally doped AlGaIn/GaN multichannel high-electron-mobility transistor heterostructures—Experimental evidence of the hole trap state,” *Appl. Phys. Lett.* **123**, 122105 (2023).
- ⁹A. Y. Polyakov *et al.*, “Effects of InAlN underlayer on deep traps detected in near-UV InGaIn/GaN single quantum well light-emitting diodes,” *J. Appl. Phys.* **126**, 125708 (2019).
- ¹⁰C. G. Van de Walle and J. Neugebauer, “First-principles calculations for defects and impurities: Applications to III-nitrides,” *J. Appl. Phys.* **95**, 3851 (2004).
- ¹¹M. A. Reshchikov and H. Morozov, “Luminescence properties of defects in GaN,” *J. Appl. Phys.* **97**(6), 061301 (2005).
- ¹²C. A. Schaake *et al.*, “A donor-like trap at the InGaIn/GaN interface with net negative polarization and its possible consequence on internal quantum efficiency,” *Semicond. Sci. Technol.* **28**, 105021 (2013).
- ¹³Z. Wang *et al.*, “Characterization study of deep-level defect spatial distribution, emission mechanisms, and structural identification,” *J. Phys. D* **58**, 032501 (2025).
- ¹⁴V. I. Zubkov, I. V. Ivanova, and M. Weyers, “Direct observation of resonant tunneling in heterostructure with a single quantum well,” *Appl. Phys. Lett.* **119**(4), 043503 (2021).
- ¹⁵Z. Wang *et al.*, “Characterizing defects in p-n junctions: An analysis of admittance spectroscopy,” *J. Phys. D* **58**, 432501 (2025).
- ¹⁶G. Vincent, D. Bois, and P. Pinard, “Conductance and capacitance studies in GaP Schottky barriers,” *J. Appl. Phys.* **46**(12), 5173–5178 (1975).
- ¹⁷J. Barbollo, S. Dueñas, and L. Bailón, “Admittance spectroscopy in junctions,” *Solid-State Electron.* **35**(3), 285–297 (1992).
- ¹⁸D. L. Losee, “Admittance spectroscopy of impurity levels in Schottky barriers,” *J. Appl. Phys.* **46**(5), 2204–2214 (1975).
- ¹⁹B. Lu *et al.*, “Trap-assisted tunneling as possible carrier escape mechanism in InGaIn/GaN light-emitting diodes,” *J. Appl. Phys.* **132**, 174505 (2022).
- ²⁰E. M. Bourim and J. I. Han, “Electrical characterization and thermal admittance spectroscopy analysis of InGaIn/GaN MQW blue LED structure,” *Electron. Mater. Lett.* **11**(6), 982–992 (2015).
- ²¹F. Luc, E. Rosencher, and B. Vinter, “Determination of electron recombination parameters in GaAs/AlGaAs quantum wells by impedance spectroscopy,” *Appl. Phys. Lett.* **62**(10), 1143–1145 (1993).